

Pixel Capture User guide



Shot on Pixel Capture

Created in 2024

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Shot on Pixel Capture

Intro

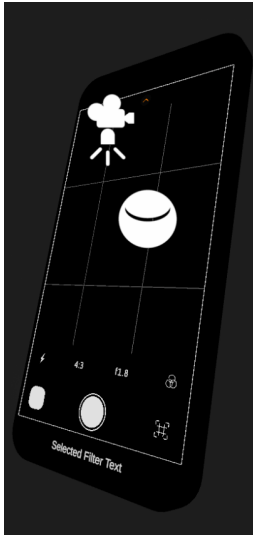
Welcome to Pixel Capture! This Unity asset lets you add a fun photo mode mechanic to your game. Whether you want to snap pictures using a phone-style interface on the UI Canvas or directly on a physical phone, Pixel Capture has got you covered.

With Pixel Capture, you can capture moments in your game from a unique perspective, perfect for adventure games, horror games or any other application. This documentation will help you set it up and start using it quickly and easily. Check out the table of contents above to get started!

Prefabs

Prefab directory: **Assets/Detail Studios/Pixel Capture/Prefabs**

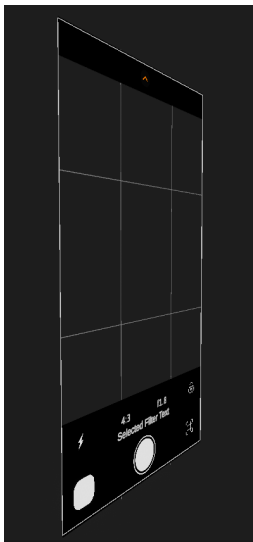
To suit the needs of your project there are two prefabs to start out with:



Handheld Phone

Description: A physical phone object featuring the same user interface as the Fullscreen UI but set to world space for interaction in 3D. It includes a dedicated flash and camera displayed on the phone's screen, making it ideal for first-person view games like exploration or horror genres.

Preview: Play the **FirstPersonDemo**, for more info view the **Demo** chapter



Fullscreen UI

Overlay

Description: A camera user interface that fills the entire screen. Filters are applied directly to the main camera instead of a separate "phone camera." This traditional photo mode uses the main camera and a 2D user interface for interaction, ideal for seamless photo capturing within the game's main view.

Preview: Play the **FullscreenUIDemo**, for more info view the **Demo** chapter

Setting up a new project

Setting up Pixel Capture in a blank scene is easy:

Note: This asset requires the post processing package to run

Step 1: Add one of the two prefabs located here in your project: **Assets/Detail Studios/Pixel Capture/Prefabs**. For help deciding which prefab to use, please consult the **Prefabs** section. Depending on the prefab you selected, use the following instructions:

Fullscreen Capture UI

Step 2: You should now see a large user interface covering the “Game” window. If you currently have a camera in the scene, it is recommended that you replace it with the “PhoneCamera” object located in the Fullscreen UI **Prefabs**. This is necessary because the main camera is used for filters and click detection. If you’re having issues check out the **Troubleshooting** section.

Handheld Phone

Step 2: Reset the location of the **Prefabs** by selecting the object, right-clicking on the Transform component, and choosing "Reset Position." The location of the reset button may vary depending on the Unity version.

Step 3: With the phone now centered in the scene, you can start using it. The program should auto-detect the main camera and use it for click detection. If you can’t click on your canvas, please consult the **Troubleshooting** section.

Demo

The pixel capture asset includes three demos located here in your project: **Assets/Detail Studios/Pixel Capture/Demo**

Static Handheld Demo



Demo showing a beautiful abstract scene with a static viewpoint providing consistent images great for testing the settings.

First Person Demo



The first-person demo really tests the autofocus to its limits and is a great showcase of an application in an adventure game

Fullscreen UI



The Fullscreen UI is a barebones showcase of the user interface consisting of only the canvas and a camera

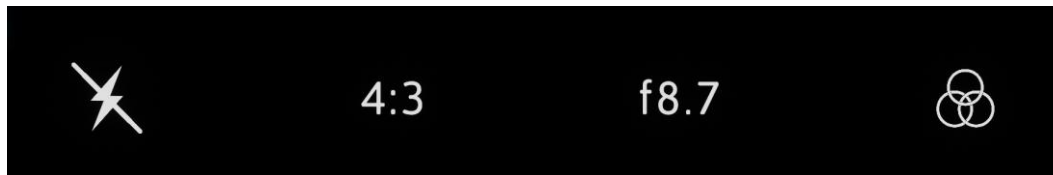
User Interface



The interface is simple and easy to understand. We have taken inspiration from the classic phone camera interface. The canvas uses the unity animation system, animation clips can be found here:

(ProjectDirectory)/ Assets/Detail Studios/Pixel Capture/animations

Pressing the expand button at the top  will expand the additional settings menu



Flash

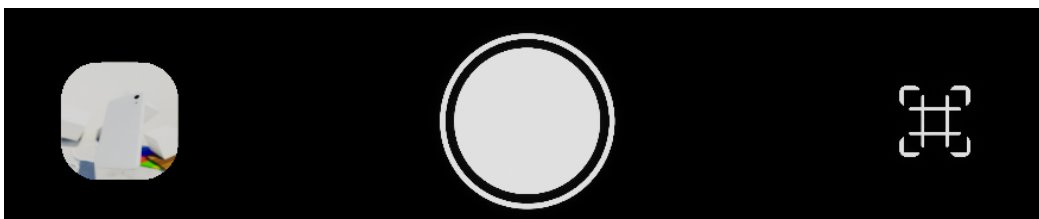
Aspect ratio

F-stop

Filters

Here you can activate the flash, change the aspect ratio, adjust the aperture and experiment with different filters.

Looking towards the bottom of the UI you can find three buttons:



Open image library

Take a photo

Activate Grid

Just like any camera app you know, there is a shutter button in the centre and a button leading to the image library to the left. However, unlike other apps you have a grid button to the right activating the camera grid.

Important Directories

Here's where you can find key components of Pixel Capture within your project directory:

- **Pixel Capture Images (JPG):**
 - (ProjectDirectory)/Assets/Detail Studios/Pixel Capture/Resources/RenderOutput
- **Image Filters (Post Process Profile):**
 - (ProjectDirectory)/Assets/Detail Studios/Pixel Capture//Resources/Color Filters
- **Demo Scenes (Scene Asset):**
 - (ProjectDirectory) /Assets/ Detail Studios/Pixel Capture//Demo
- **Camera Preview Render Texture:**
 - Located here, you can control the settings of the live preview in the Handheld Phone **Prefabs**.
 - (ProjectDirectory)/ Assets/ Detail Studios/Pixel Capture//CameraRenderTexture

Adding new filters

Note if you already know how to create a post processing profile and a color grading component to it, please skip to step 3 “Use the Custom Editor Menu to Add the Profile:”

1. Create a New Post Process Profile:

1. Navigate to the Asset Menu:

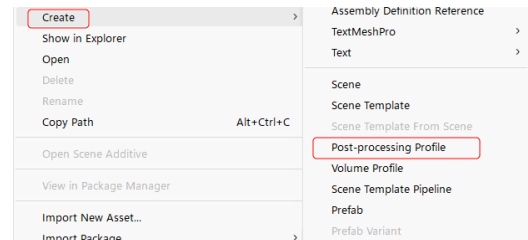
- Click on Assets in the top menu.

2. Create a New Post Process Profile:

- Hover over Create in the dropdown menu.
- Select Post-Processing Profile.

3. Name Your Profile:

- Rename the newly created profile as desired to whatever you want (e.g., "MyColorGradeProfile").



2. Add a Color Grading Component:

1. Select the Post Process Profile:

- In the Project window, select the newly created Post Process Profile (MyColorGradeProfile).

2. Navigate to the Inspector:

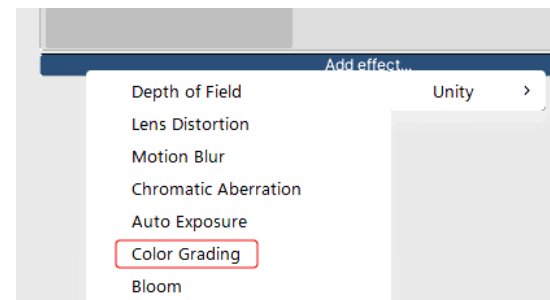
- Scroll down in the Inspector until you find the Add Component button.

3. Add Color Grading Component:

- Click Add Component.
- Type Color Grading in the search bar.
- Select Color Grading from the dropdown menu.

4. Configure Color Grading Settings:

- Adjust the settings of the Color Grading component as needed to achieve the desired look for your project.



3. Use the Custom Editor Menu to Add the Profile:

1. Open the Pixel Capture Menu:

- In the Unity Editor, click on Pixel Capture in the top toolbar.

2. Select "Add New Filter":

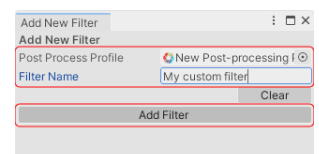
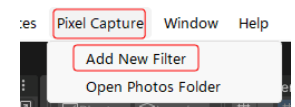
- Click on Add New Filter from the dropdown menu.

3. Enter Profile Details:

- In the opened window, select the newly created PostProcessProfile using the object field.
- Enter a name for the filter in the Filter Name input field.

4. Add the Filter:

- Click Add Filter to add the Color Grading profile to your project



Troubleshooting

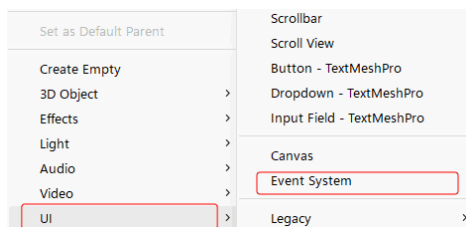
Having issues? Here are some common problems with pixel capture

My canvas isn't responding to my mouse input

This is usually due to one of two reasons:

There is no event system in the scene

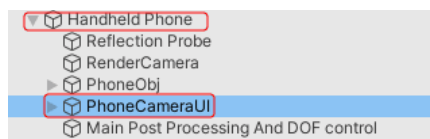
The Fullscreen UI **Prefabs** and the Handheld phone **Prefabs** both include event systems however for whatever reason your project may not have one. To add an event system, right click in the hierarchy then select UI->Event system



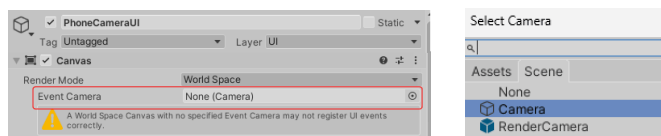
The event camera is not set

The Handheld Phone uses a world space canvas that requires that you select a event camera to process mouse inputs. If it is not set the pixel capture script will automatically set the event camera to any existing camera that isn't the phone rendering camera. If this camera is incorrect, you may not be able to click on your canvas.

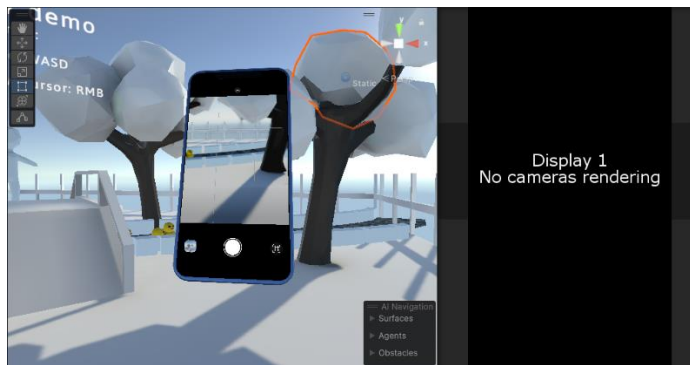
To fix this select the Handheld Phone **Prefabs**, then select the PhoneCameraUI object.



In the inspector change the event camera on the canvas component to the camera that you're using as your main camera.



No cameras rendering

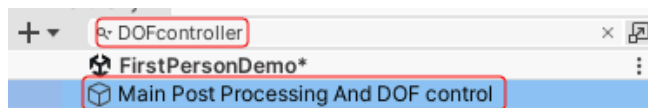


If you're seeing this, you need to add another camera to the scene. The camera included with the Handheld Phone **Prefabs** is only used to render to the display for a live preview. You also need another camera in the scene to see the phone. To add another camera simply right click in the hierarchy and left click on camera. You should now be seeing an image in the game window

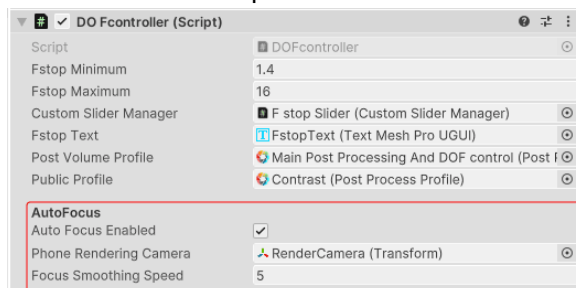
The camera autofocus isn't working

Note: Since the autofocus uses a raycast system every object you want to focus on needs a collider

If your encountering this issue you may need to edit the parameters of the DOFcontroller component(Depth of field controller). To find this component simply use the hierarchy search and search for "DOFcontroller" and select the "Main Post Processing And DOF control" object.



Now in the inspector scroll to the bottom and ensure that the settings of the DOFcontroller component are correct

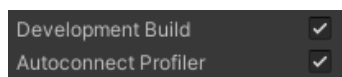


The autofocus should be set to enabled and the phone rendering camera should be set to the transform of the camera attached to the handheld phone **Prefabs**

Low Fps in build

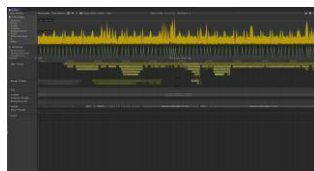
1. Enable Development Build and Autoconnect Profiler

1. **Open Build Settings:** Navigate to File -> Build Settings.
2. **Enable Development Build:** Check the Development Build option.
3. **Autoconnect Profiler:** Check the Autoconnect Profiler option.

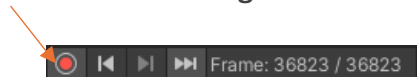


2. Run and Profile Your Project

1. **Build and Run:** Click Build and Run in the Build Settings window.
2. **Profile Window:** After the project runs, a large profiler window should appear.



3. **Start Recording:** Press the record button in the profiler window.



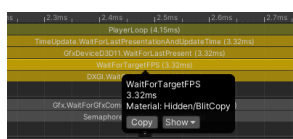
3. Analyse the Timeline

1. **Open Timeline:** In the profiler, open the timeline view.
2. **Identify Bottlenecks:** Look for events that take longer than 1ms. These events are potential performance bottlenecks.



4. Check for Vsync Issues

1. **Identify Vsync Delays:** Look for WaitForTargetFPS or WaitForLastPresentationAndUpdateTime in the timeline. If WaitForTargetFPS takes 3-5ms, Vsync might be causing the slowdown.



5. Disable Vsync

1. **Open Project Settings:** Navigate to Edit -> Project Settings.
2. **Quality Settings:** Select Quality in the Project Settings window.
3. **Set VSync Count:** Change the VSync Count setting to Don't Sync.

Any other issues?

Contact me!

Contact

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